



With the size of an organisation's infrastructure increasing, monitoring is becoming a challenge. Ganglia presents itself as a very good solution when it comes to cluster-based monitoring, and analysing the available data.

Monitoring multiple data centres is becoming a daunting task for sysadmins, and it certainly starts to become tiresome as the number of nodes increase rapidly. Systems administrators would typically want to check systems statistics like the disk, CPU and network utilisation, etc. They would be interested in knowing whether systems are performing within thresholds, where the bottlenecks are occurring, and how applications are performing. Now, when managing a few systems, you could probably log in to each system and check—but, as mentioned earlier, it becomes a problem as infrastructure grows.

So, you need a performance monitoring system to solve your problem—one that does not overload you with a lot of unwanted statistics, but only those that actually help you identify problems. With these conditions in mind, Ganglia is surely an option to explore. The good thing is that once you have it running, you can customise it to a great extent, writing your own custom metrics.

Ganglia is an open source project, originally designed by the University of California. It acts as a distributed monitoring system for high-performance systems like clusters and grids. For a more formal definition: "Ganglia is a scalable distributed monitoring system for high-performance computing systems such as clusters and grids. It is based on a hierarchical design, targeted at federations of clusters. It leverages widely used technologies such as XML for data representation, XDR for compact, portable data transport, and RRDtool for data storage and visualisation." (from <http://ganglia.info/>)

Here is a small glossary of terms used in Ganglia:

- Host/node: Typically, a machine.
- Cluster: A group of nodes.
- Grid: A group of clusters.
- Metrics: The graphs that are displayed.
- RRDs or Round Robin Databases: According to its website, RRDtool (the Round Robin Database tool) is a system to store and display time-series data (e.g., network bandwidth, machine-room temperature, server load average, etc). It stores the data very compactly to maintain a manageable archive size. RRDtool presents useful graphs (see Figure 1) by processing the data to enforce a certain data density.

Ganglia architecture

The Ganglia architecture (see Figure 2) has the following main components:

- Ganglia monitoring daemon (*gmond*)
- Ganglia meta daemon (*gmetad*)
- RRD
- Ganglia front-end

Ganglia monitoring daemon

gmond is installed on all nodes that need to be monitored. It monitors the changes in host state, listens to other *gmond* instances over multi-cast/unicast, and is responsible for sending XML, over a TCP connection, to *gmetad*. To get started, you need to edit the cluster information in the configuration file. Given below is a sample of a *gmond.conf* configuration file to get started with: